Azure Storage Service — Brief Instructor Overview

Azure Storage is a highly scalable, durable, secure, and cost-effective cloud storage solution provided by Microsoft Azure. It is designed to store massive amounts of structured, semi-structured, and unstructured data and is widely used by cloud-native and enterprise applications.

Core Azure Storage Services

1. Azure Blob Storage

- Used for storing **unstructured data** such as images, videos, documents, backups, and logs.
- Data is stored as **blobs** inside **containers**.
- Common use cases:
  - File uploads/downloads
  - Media streaming
  - Application logs and backups
  - Data lake scenarios

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2. Azure File Storage

- Provides **fully managed file shares** in the cloud.
- Accessible using standard protocols like **SMB** and **NFS**.
- Common use cases:
  - Lift-and-shift of legacy applications
  - Shared file systems between VMs
  - Replacing on-prem file servers

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3. Azure Queue Storage

- A messaging service for storing and retrieving messages.

- Enables **asynchronous communication** between application components.
- Common use cases:
    - Background processing
    - Decoupling microservices
    - Task scheduling

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4. Azure Table Storage

- A **NoSQL key-value** storage service.
- Schema-less design with fast access using **PartitionKey** and **RowKey**.
- Common use cases:
    - Storing large volumes of simple structured data
    - User profiles, configuration data
    - Metadata storage

Key Features (Why Azure Storage?)

- **High Durability**: Multiple replicas of data (LRS, ZRS, GRS, RA-GRS)
- **Scalability**: Scales automatically to petabytes of data
- **Security**: Encryption at rest and in transit, RBAC, SAS tokens
- **Cost-Effective**: Pay only for what you use
- **High Availability**: 99.9%+ SLA depending on configuration

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Common Real-World Scenarios

- ASP.NET Core applications storing user-uploaded files
- Microservices exchanging messages using queues
- Backup and disaster recovery solutions
- Log and telemetry storage for cloud applications

One-Line Summary

Azure Storage is the foundational cloud storage platform in Azure that supports files, objects, queues, and NoSQL data with enterprise-grade security and scalability.

Creating an Azure Storage Account for Blob Storage and Queue Storage

(Step-by-step Instructor Guide)

In Azure, **Blob Storage** and **Queue Storage** are not created separately. Both are enabled automatically inside a **single Storage Account**.

Prerequisites

- An active Azure subscription
- Access to **Azure Portal**

Step 1: Sign in to Azure Portal

1. Open the Azure Portal.
2. Log in using your Microsoft account credentials.

Step 2: Create a Storage Account

1. In the search bar, type **Storage accounts**
2. Click **Storage accounts**
3. Click **Create**

Step 3: Basics Tab Configuration

1. Subscription

- Select your Azure subscription

2. Resource Group

- Click **Create new**

- Example: rg-storage-demo

3. Storage Account Name

- Must be **globally unique**
- Only lowercase letters and numbers
- Example: demostorageacct123

4. Region

- Choose the nearest region (e.g., Central India)

5. Performance

- Select **Standard**
    - (Recommended for Blob & Queue demos)

6. Redundancy

- Choose **LRS (Locally Redundant Storage)** for demo/practice
- Production may use GRS or ZRS

Click **Next: Advanced**

Step 4: Advanced Tab

- Keep defaults
- Ensure **Blob public access** is enabled (optional, for learning)
- Click **Next: Networking**

Step 5: Networking Tab

- Select **Public endpoint (all networks)** for learning
- Click **Next: Data protection**

Step 6: Data Protection Tab

- Keep defaults

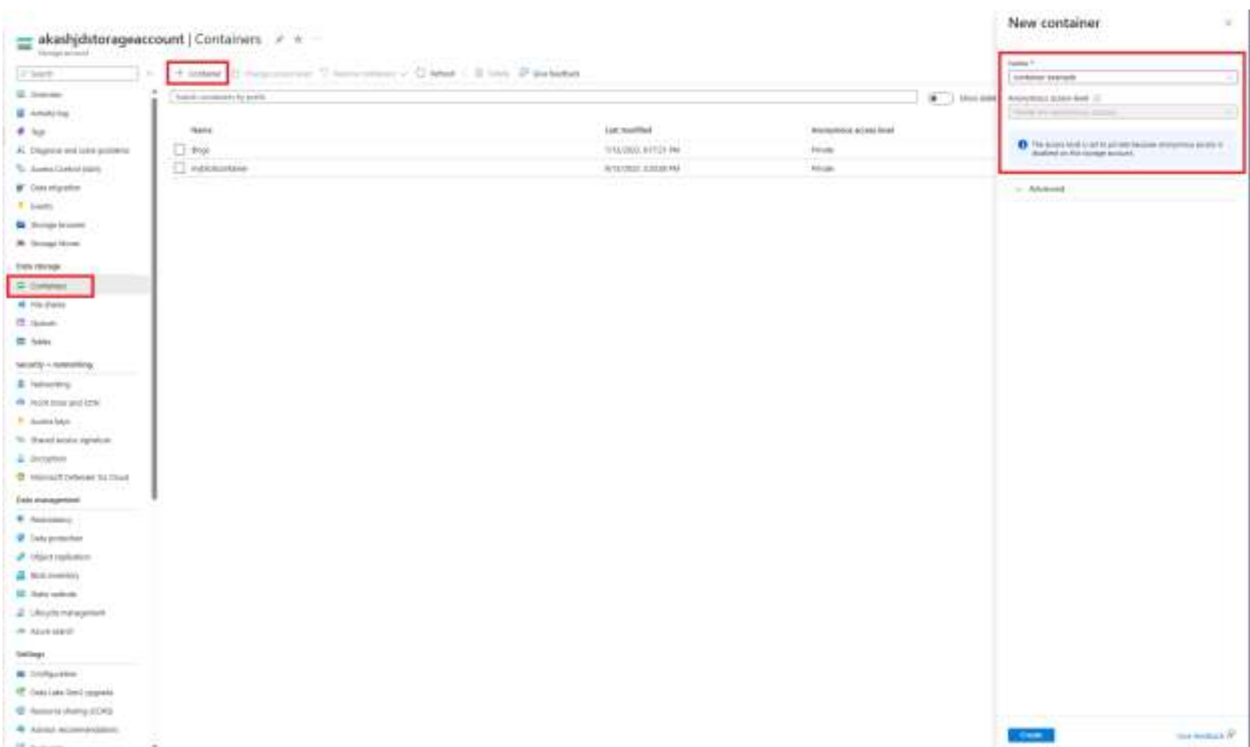
- Click **Next: Review + Create**

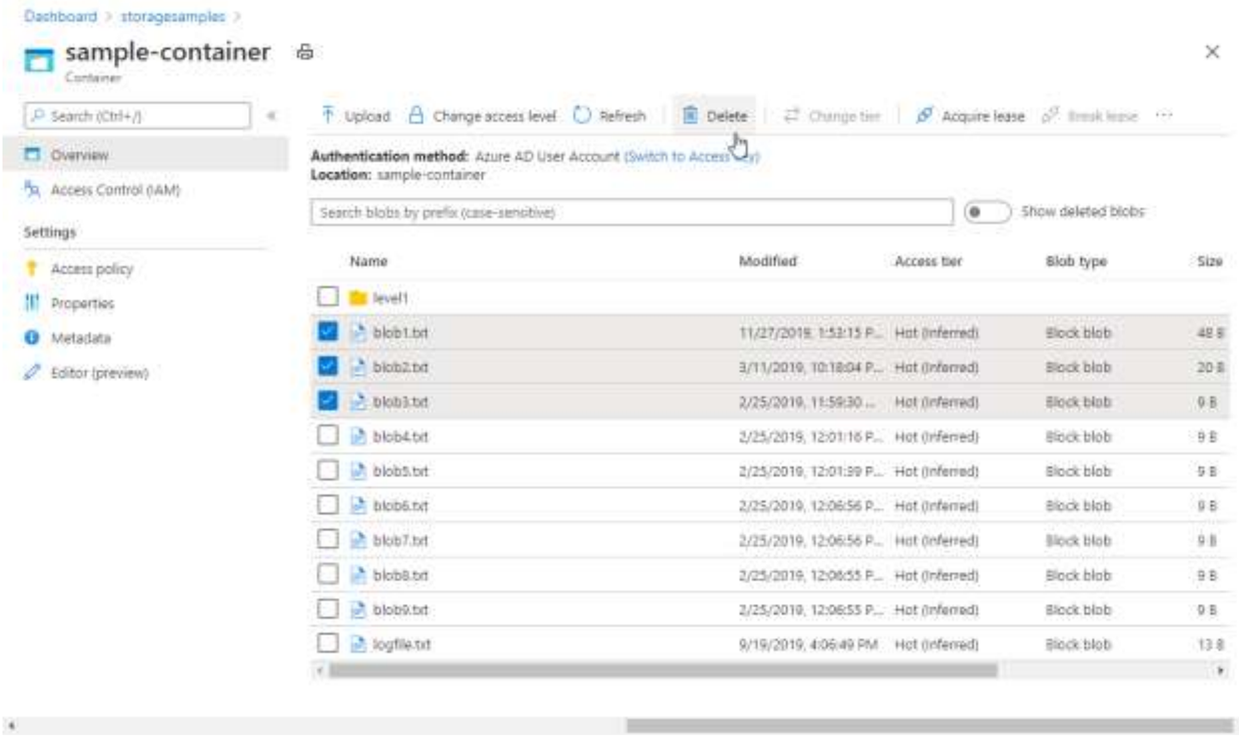
Step 7: Review and Create

1. Review all settings
2. Click **Create**
3. Wait for deployment to complete
4. Click **Go to resource**

Your **Storage Account** is now created.

Creating Blob Storage (Container)





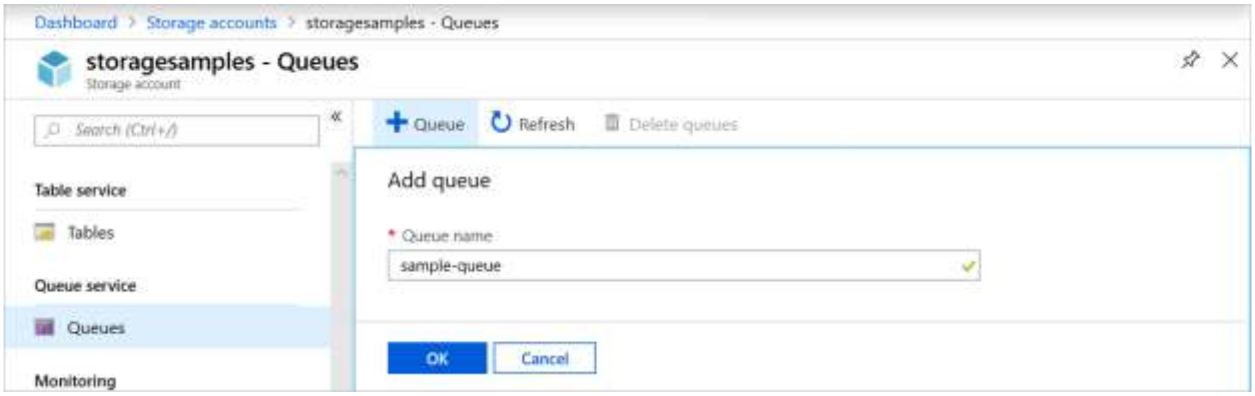
Step 8: Create a Blob Container

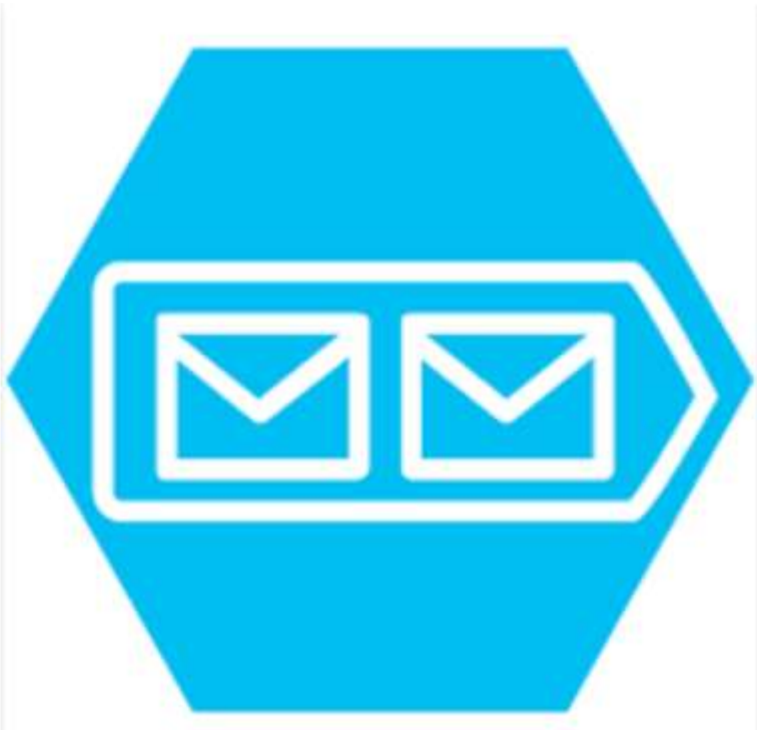
1. Open your **Storage Account**
2. In the left menu, select **Containers** (under Data storage)
3. Click **+ Container**
4. Enter:
   - Name: images or documents
   - Public access level: Private
5. Click **Create**

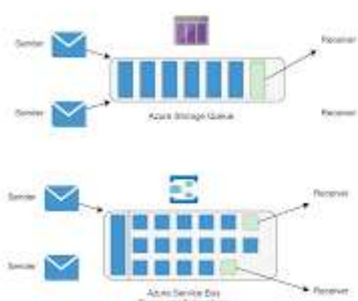
Upload a Blob

1. Open the container
2. Click **Upload**
3. Select a file
4. Click **Upload**

Creating Queue Storage (Queue)







Step 9: Create a Queue

1. In the same **Storage Account**
2. Select **Queues** (under Data storage)
3. Click **+ Queue**
4. Enter Queue Name:
    - Example: order-processing-queue
5. Click **Create**

Your Queue Storage is ready.

Step 10: Access Keys & Connection String

1. Go to **Access keys**
2. Copy:
    - **Connection string**
    - **Account key**
3. Use this in:
    - ASP.NET Core
    - Azure Functions
    - Console apps

Architecture Summary

Component	Purpose
Storage Account	Parent resource
Blob Container	Stores files (images, documents, videos)
Queue	Stores messages for async processing
Access Keys	Authentication

Typical Use Case (Real-World)

- **Blob Storage** → Upload invoices or images
- **Queue Storage** → Send message for background processing
- Azure Function / Worker Service processes the queue message

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One-Line Summary

A **single Azure Storage Account** hosts **Blob Storage for files** and **Queue Storage for messaging**, both created and managed independently inside the account.